DATA MINING AND WAREHOUSING

Complete Unit Notes – Diploma Engineering (GTU)

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Prepared in simple English for beginner Diploma Engineering students. Each topic is structured with Definition/Explanation, 3 Features, 3 Uses, Applications, and a Real-Life Example.

UNIT 1 – Fundamental Concepts of Data Mining

1.1 Data Mining – Definition

Definition / Explanation

Data mining is the process of finding useful patterns, relationships, trends, and knowledge from a large amount of data. It uses different techniques to examine data and discover information that may not be easy to see directly. Data mining helps organizations understand past data and use it for better decisions.

Features

- Find useful patterns in large datasets.
- Identify trends and relationships in data.
- Support prediction and decision-making.

Uses

- Business analysis
- Customer purchase analysis
- Student performance analysis

Applications

- Business analysis
  - Customer purchase analysis
  - Student performance analysis

Real-Life Example

Real-life example: An online shopping website studies previous purchases to find which products customers often buy together.

1.1.2 Data Mining and Database

Definition / Explanation

A database is mainly used to store, organize, and retrieve data, while data mining is used to analyze stored data and discover useful patterns or knowledge. A database can answer direct queries such as finding a student's marks. Data mining can examine many student records and discover patterns such as factors related to good performance. Thus, database focuses on data storage and retrieval, whereas data mining focuses on discovering knowledge from data.

Features

- Database stores organized data.
- Data mining analyzes existing data.
- Both can work together to support useful information.

Uses

- Storing records
- Searching records
- Discovering patterns from stored records

Applications

- Storing records
  - Searching records
  - Discovering patterns from stored records

Real-Life Example

Real-life example: A college database stores student marks, while data mining can analyze those marks to identify performance trends.

1.1.3 Patterns, Trends and Predictions

Definition / Explanation

A pattern is a repeated relationship or behavior found in data. A trend shows how data changes over time or in a particular direction. A prediction is an estimate of a future result based on available data and discovered patterns. These three concepts are commonly used in data mining to convert large datasets into useful knowledge. For example, purchase records may show a pattern between two products, a trend of increasing sales, and a prediction of future demand.

Features

- Patterns show repeated relationships.
- Trends show changes or directions.
- Predictions estimate possible future results.

Uses

- Sales analysis
- Customer behavior analysis
- Demand forecasting

Applications

- Sales analysis
  - Customer behavior analysis
  - Demand forecasting

Real-Life Example

Real-life example: A shop finds that umbrella sales increase during rainy months and predicts higher umbrella demand during the next rainy season.

1.2 Steps in Knowledge Discovery Process (KDD)

Definition / Explanation

Knowledge Discovery in Databases (KDD) is a process used to obtain useful knowledge from data. The main steps in the syllabus are data selection, data preprocessing, data mining, and interpretation and evaluation of results. First, required data is selected. Next, incorrect, missing, or unnecessary data is prepared. Then a data mining technique is applied to find patterns. Finally, the results are interpreted and evaluated to decide whether they are useful.

Features

- Select relevant data.
- Preprocess the selected data.
- Mine and evaluate useful patterns.

Uses

- Student analysis
- Business analysis
- Healthcare data analysis

Applications

- Student analysis
  - Business analysis
  - Healthcare data analysis

Real-Life Example

Real-life example: A college selects student marks, cleans incorrect entries, analyzes performance patterns, and evaluates the results.

1.2.1 Data Selection, Data Preprocessing and Data Mining

Definition / Explanation

Data selection means choosing the required data from a larger collection. Data preprocessing means preparing the selected data so that it is suitable for analysis. It may involve handling missing, incorrect, or unnecessary values. Data mining is then applied to the prepared data to discover useful patterns or knowledge. These steps are important because poor-quality or irrelevant data can affect the results of data mining.

Features

- Select relevant records and attributes.
- Clean and prepare the selected data.
- Apply a suitable mining technique.

Uses

- Preparing student datasets
- Preparing sales records
- Preparing patient datasets

Applications

- Preparing student datasets
  - Preparing sales records
  - Preparing patient datasets

Real-Life Example

Real-life example: Before analyzing student results, a college selects required marks, removes incorrect entries, and then analyzes the cleaned data.

1.2.2 Interpretation and Evaluation of Results

Definition / Explanation

After data mining produces patterns or results, they must be interpreted and evaluated. Interpretation means understanding what the discovered result means in the real situation. Evaluation checks whether the result is useful, meaningful, and suitable for the purpose of analysis. Results that are not useful may be rejected or require further analysis. This step helps convert technical mining results into information that people can understand and use.

Features

- Understand the discovered results.
- Check whether results are meaningful.
- Select useful results for decision-making.

Uses

- Business decisions
- Student performance review
- Healthcare analysis

Applications

- Business decisions
  - Student performance review
  - Healthcare analysis

Real-Life Example

Real-life example: If analysis shows that attendance is related to student performance, a college can interpret and evaluate whether the result is strong enough to support action.

1.3 Importance of Data Mining

Definition / Explanation

Data mining is important because large amounts of data are generated and stored every day. Simply storing data is not enough; useful knowledge must be discovered from it. Data mining helps identify patterns, trends, and relationships that can support better decisions. It can be used to understand customer behavior, analyze student performance, predict diseases, and detect fraud. It therefore helps organizations use their existing data more effectively.

Features

- Helps discover useful knowledge.
- Supports better decision-making.
- Finds patterns and trends in large datasets.

Uses

- Business analysis
- Fraud detection
- Performance analysis

Applications

- Business analysis
  - Fraud detection
  - Performance analysis

Real-Life Example

Real-life example: A bank can analyze transaction data to identify unusual patterns that may indicate fraud.

1.4 Applications of Data Mining in Real Life

Definition / Explanation

Data mining is used in many real-life areas. In education, it can analyze student performance. In business, it can study customer purchases and behavior. In healthcare, it can support disease prediction and analysis of patient data. In banking, it can help detect suspicious transaction patterns and fraud. These applications show how data mining can convert stored data into useful information for practical decisions.

Features

- Education – student performance analysis.
- Business – customer purchase analysis.
- Healthcare and banking – disease prediction and fraud detection.

Uses

- Schools and colleges
- Retail and business organizations
- Hospitals and banks

Applications

- Schools and colleges
  - Retail and business organizations
  - Hospitals and banks

Real-Life Example

Real-life example: A shopping company analyzes customer purchase history to understand buying behavior and improve its decisions.

UNIT 2 – Basics of Data Warehousing

2.1 Meaning of Data, Information and Knowledge

Definition / Explanation

Data, information, and knowledge are related but different concepts. Data is a collection of raw facts, such as numbers, names, or dates. Information is processed or organized data that has meaning. Knowledge is the understanding gained by analyzing information and using it to make decisions. The three can be viewed as a flow: raw data is processed into information, and information is used to develop knowledge.

Features

- Data is raw facts.
- Information is meaningful processed data.
- Knowledge helps in understanding and decision-making.

Uses

- Student marks
- Sales reports
- Business decisions

Applications

- Student marks
  - Sales reports
  - Business decisions

Real-Life Example

Real-life example: Individual marks are data; a class result report is information; understanding which subjects need improvement is knowledge.

2.1.1 Definition and Examples of Data

Definition / Explanation

Data means raw facts or values collected for a purpose. It may contain numbers, text, dates, names, measurements, or other recorded values. By itself, data may not provide a complete meaning. It becomes more useful after it is organized and processed. In data mining and warehousing, large amounts of data are collected from different sources and later analyzed to obtain useful information.

Features

- Can contain numbers, text, or dates.
- Represents recorded facts.
- Can be stored for later processing and analysis.

Uses

- Student marks
- Product prices
- Customer names

Applications

- Student marks
  - Product prices
  - Customer names

Real-Life Example

Real-life example: The marks 72, 81, and 65 recorded for three students are raw data.

2.1.2 How Data Becomes Information

Definition / Explanation

Data becomes information when it is processed, organized, and given meaning. Raw values may be difficult to understand when viewed separately. When they are arranged into tables, reports, summaries, or calculations, they can provide useful meaning. Processing may include sorting, calculating totals, grouping records, or generating reports. Information helps people understand what the data represents and supports further analysis and decision-making.

Features

- Organize raw data.
- Process or summarize it.
- Give the result a useful meaning.

Uses

- Reports
- Charts
- Summary tables

Applications

- Reports
  - Charts
  - Summary tables

Real-Life Example

Real-life example: Individual monthly sales values become information when they are totaled to show the company's total monthly sales.

2.1.3 What is Knowledge

Definition / Explanation

Knowledge is the useful understanding obtained from information, analysis, and experience. It helps a person or organization understand a situation and decide what action may be appropriate. In data mining, discovered patterns and meaningful information can contribute to knowledge. Knowledge is more useful than raw data because it provides an understanding that can guide practical decisions.

Features

- Comes from meaningful information.
- Provides understanding of a situation.
- Supports decisions and actions.

Uses

- Business planning
- Student improvement
- Customer behavior analysis

Applications

- Business planning
  - Student improvement
  - Customer behavior analysis

Real-Life Example

Real-life example: A report shows low sales of a product, and the company understands that demand is falling and decides to review its marketing.

2.1.4 Real-Life Examples of Data, Information and Knowledge

Definition / Explanation

A simple example can show the difference between data, information, and knowledge. Suppose a shop records daily sales amounts. The individual sales values are data. When these values are organized into a weekly report showing total and average sales, they become information. When the shop understands that sales are higher on weekends and uses this understanding for planning, it becomes knowledge. This flow is useful in many real-life systems.

Features

- Data records raw facts.
- Information gives processed meaning.
- Knowledge supports understanding and action.

Uses

- Retail sales
- Student results
- Healthcare records

Applications

- Retail sales
  - Student results
  - Healthcare records

Real-Life Example

Real-life example: Student marks are data, a result sheet is information, and identifying weak subjects for improvement is knowledge.

2.2 Introduction to Data Warehouse

Definition / Explanation

A data warehouse is a centralized storage system designed to collect and organize data from different sources for analysis and reporting. It is mainly used to support decision-making rather than day-to-day transaction processing. A data warehouse can bring together historical data from an organization so that users can analyze trends and prepare reports. It provides a useful base for business intelligence and data analysis.

Features

- Collects data from different sources.
- Stores data for analysis and reporting.
- Supports decision-making.

Uses

- Business reporting
- Trend analysis
- Management analysis

Applications

- Business reporting
  - Trend analysis
  - Management analysis

Real-Life Example

Real-life example: A retail company can store sales information from many branches in a data warehouse and analyze overall sales trends.

2.2.2 Key Features of a Data Warehouse

Definition / Explanation

A data warehouse has features that make it suitable for analysis. It collects data from different sources and keeps it organized for analytical use. It can store historical data so that changes can be studied over time. Data is prepared and integrated before being used for reporting. The warehouse therefore provides a central place for organizations to analyze information and support decision-making.

Features

- Integrates data from different sources.
- Stores historical data.
- Supports analysis and reporting.

Uses

- Trend analysis
- Management reporting
- Business intelligence

Applications

- Trend analysis
  - Management reporting
  - Business intelligence

Real-Life Example

Real-life example: A company can keep several years of sales data in a warehouse to compare yearly performance.

2.2.3 Database and Data Warehouse

Definition / Explanation

A database is generally designed to store and manage current operational data and support regular transactions. A data warehouse is designed mainly for analysis, reporting, and decision-making. Databases are commonly used for activities such as inserting, updating, and retrieving individual records. Data warehouses bring together data, often including historical data, so larger analytical queries can be performed. Both store data, but their main purposes are different.

Features

- Database focuses on operational data.
- Data warehouse focuses on analysis.
- Both organize and store data for useful access.

Uses

- Transaction processing
- Reporting
- Historical analysis

Applications

- Transaction processing
  - Reporting
  - Historical analysis

Real-Life Example

Real-life example: A college database stores current student records, while a data warehouse can combine historical results to analyze performance trends.

2.2.4 Importance in Decision-Making

Definition / Explanation

A data warehouse supports decision-making by providing organized and integrated data for analysis. Managers can use reports and historical information to understand trends and compare performance.

Instead of examining data from many separate systems, users can analyze information from a centralized warehouse. This can help organizations understand what has happened and identify useful patterns for planning and decision-making.

Features

- Provides integrated information.
- Supports historical analysis.
- Helps managers make informed decisions.

Uses

- Sales planning
- Performance comparison
- Business reporting

Applications

- Sales planning
  - Performance comparison
  - Business reporting

Real-Life Example

Real-life example: A company compares sales from several years in its data warehouse to plan future business activities.

2.3 OLTP and OLAP

Definition / Explanation

OLTP stands for Online Transaction Processing. It is used for regular day-to-day transactions such as inserting, updating, and retrieving operational records. OLAP stands for Online Analytical Processing. It is used to analyze data from different views and support decision-making. OLTP focuses mainly on transaction processing, while OLAP focuses on analysis. Both are important in data systems, but they serve different purposes.

Features

- OLTP handles day-to-day transactions.
- OLAP performs analytical queries.
- Both process organizational data for different purposes.

Uses

- OLTP: banking transactions
- OLTP: order entry
- OLAP: sales analysis

Applications

- OLTP: banking transactions
  - OLTP: order entry
  - OLAP: sales analysis

Real-Life Example

Real-life example: A bank uses OLTP to record a money transfer, while management can use OLAP to analyze monthly transaction trends.

2.3.2 Major Differences and Examples of OLTP and OLAP

Definition / Explanation

OLTP and OLAP differ mainly in their purpose. OLTP is designed for frequent operational transactions, while OLAP is designed for analysis and reporting. OLTP commonly works with current operational data and individual transactions. OLAP can analyze large amounts of historical data and provide summaries from

different views. For example, an online shopping system uses OLTP to record an order, while OLAP can analyze yearly sales by product and region.

Features

- OLTP supports transaction processing.
- OLAP supports analytical processing.
- Their data use and query purposes are different.

Uses

- Order processing
- Banking transactions
- Sales and trend analysis

Applications

- Order processing
  - Banking transactions
  - Sales and trend analysis

Real-Life Example

Real-life example: Recording a customer's purchase is OLTP; analyzing total purchases by month and product is OLAP.

2.4 Data Marts

Definition / Explanation

A data mart is a smaller, focused collection of data designed for a particular department, subject, or business area. It can provide users with relevant data without requiring them to work with the complete organizational data warehouse. For example, a sales data mart may contain information mainly required by the sales department. Data marts are useful for focused analysis and reporting.

Features

- Focuses on a specific business area.
- Contains relevant data for selected users.
- Supports focused analysis and reporting.

Uses

- Sales data mart
- Finance data mart
- Marketing data mart

Applications

- Sales data mart
  - Finance data mart
  - Marketing data mart

Real-Life Example

Real-life example: A company's sales department can use a sales data mart to analyze sales information without working with unrelated company data.

2.4.2 Data Warehouse and Data Mart

Definition / Explanation

A data warehouse generally contains integrated data for the organization, while a data mart focuses on a particular department or business area. A data mart may be created from a data warehouse or another suitable source. The warehouse supports broader analysis, whereas the data mart provides a more focused view. Both are used to organize data for reporting and analysis.

Features

- Warehouse has broader organizational scope.
- Data mart has a focused departmental scope.
- Both support analysis and reporting.

Uses

- Enterprise reporting
- Departmental reporting
- Focused business analysis

Applications

- Enterprise reporting
  - Departmental reporting
  - Focused business analysis

Real-Life Example

Real-life example: A company-wide warehouse can contain all sales and finance data, while a sales data mart contains data mainly needed by the sales team.

2.4.3 Types of Data Marts

Definition / Explanation

Data marts can be classified according to how their data is obtained. A dependent data mart gets data from an existing data warehouse. An independent data mart gets data directly from operational or other source systems without depending on a central warehouse. A hybrid data mart can use data from more than one type of source. These types provide different ways to organize focused data for analysis.

Features

- Dependent data mart uses a data warehouse.
- Independent data mart uses source systems directly.
- Hybrid data mart can combine sources.

Uses

- Sales analysis
- Finance reporting
- Departmental analytics

Applications

- Sales analysis
  - Finance reporting
  - Departmental analytics

Real-Life Example

Real-life example: A sales department may receive its data mart from the company warehouse, making it a dependent data mart.

2.4.4 Uses of Data Marts in Organizations

Definition / Explanation

Data marts are used when a department needs focused information for analysis and reporting. They can provide relevant data to specific users and make departmental analysis easier. A sales team can use a sales data mart, while a finance team can use a finance data mart. By focusing on a particular area, users can work with the information most relevant to their responsibilities.

Features

- Support departmental analysis.

- Provide focused reporting data.
- Help users access relevant information.

Uses

- Sales reporting
- Finance analysis
- Marketing analysis

Applications

- Sales reporting
  - Finance analysis
  - Marketing analysis

Real-Life Example

Real-life example: The marketing department can use a marketing data mart to analyze customer and campaign information relevant to its work.

UNIT 3 – Data Mining Techniques

3.1 Classification

Definition / Explanation

Classification is a data mining technique used to place data into predefined classes or categories. A classification model learns from existing labeled data and then uses the learned pattern to classify new data. Common classification methods in the syllabus include Decision Tree and Naïve Bayes. Classification is useful when the possible output categories are already known, such as pass/fail or yes/no.

Features

- Uses predefined classes.
- Learns from existing labeled data.
- Classifies new records into categories.

Uses

- Student pass/fail prediction
- Disease prediction
- Customer category prediction

Applications

- Student pass/fail prediction
  - Disease prediction
  - Customer category prediction

Real-Life Example

Real-life example: A model can classify a student as likely to pass or fail using previous performance data.

3.1.2 Working of Decision Tree and Naïve Bayes

Definition / Explanation

A Decision Tree classifies data by asking a sequence of questions based on selected attributes. The branches represent conditions and the final leaves represent classes. Naïve Bayes is a classification method based on probability. It calculates the likelihood of a class using the available attributes and selects a suitable class. Both methods learn from existing data and can then classify new records.

Features

- Decision Tree uses branches and conditions.
- Naïve Bayes uses probability.
- Both classify new records using learned patterns.

Uses

- Student classification
- Email classification
- Disease prediction

Applications

- Student classification
  - Email classification
  - Disease prediction

Real-Life Example

Real-life example: A Decision Tree may ask about attendance and marks to classify a student as pass or fail.

3.1.3 Real-Life Examples of Classification

Definition / Explanation

Classification is used when data needs to be assigned to known categories. In education, it can classify students according to performance. In healthcare, it can help classify patients based on disease-related data. In business, it can classify customers into categories based on their behavior. Classification is useful because it turns existing patterns in labeled data into a model that can classify new cases.

Features

- Education – student performance classes.
- Healthcare – disease-related classes.
- Business – customer categories.

Uses

- Pass/fail prediction
- Disease prediction
- Customer classification

Applications

- Pass/fail prediction
  - Disease prediction
  - Customer classification

Real-Life Example

Real-life example: A healthcare dataset can be used to classify a new patient into a possible disease category based on symptoms and other attributes.

3.2 Clustering

Definition / Explanation

Clustering is a data mining technique that groups similar data items together without predefined class labels. Items in the same cluster are more similar to one another, while items in different clusters are less similar. Clustering is useful for discovering natural groups in data. K-means is a common clustering method included in the syllabus.

Features

- Creates groups of similar data.
- Does not require predefined class labels.
- Helps discover natural groups.

Uses

- Customer grouping
- Student grouping
- Product grouping

Applications

- Customer grouping
  - Student grouping
  - Product grouping

Real-Life Example

Real-life example: A shopping company can group customers according to similar purchasing behavior.

3.2.2 Working of K-means

Definition / Explanation

K-means is a clustering algorithm that divides data into a selected number of groups called clusters. First, the required number of clusters, K, is selected. Initial centers are chosen. Each data point is assigned to the nearest center, and the centers are recalculated using the assigned points. This assignment and recalculation process is repeated until the groups become stable or the stopping condition is reached.

Features

- Select the number of clusters K.
- Assign points to the nearest center.
- Update centers and repeat the process.

Uses

- Customer segmentation
- Student grouping
- Product grouping

Applications

- Customer segmentation
  - Student grouping
  - Product grouping

Real-Life Example

Real-life example: A store can use K-means to divide customers into three groups based on their purchasing behavior.

3.2.3 Applications of Clustering

Definition / Explanation

Clustering is useful for finding groups in data when the categories are not already known. It can group customers with similar buying behavior, students with similar learning patterns, or products with similar characteristics. The discovered groups can help organizations understand their data and plan further analysis. K-means can be applied when a suitable number of clusters is selected.

Features

- Groups similar customers or records.
- Discovers hidden groups in datasets.
- Supports analysis of group behavior.

Uses

- Customer segmentation
- Student behavior analysis
- Product grouping

Applications

- Customer segmentation
  - Student behavior analysis
  - Product grouping

Real-Life Example

Real-life example: An online store can group customers into clusters based on purchase frequency and spending behavior.

3.3 Association Rule Mining

Definition / Explanation

Association rule mining is a data mining technique used to discover relationships between items or events in a dataset. It identifies rules showing that the presence of one item may be associated with another item. It is commonly used in transaction data and market basket analysis. The Apriori algorithm is one basic method for finding association rules.

Features

- Finds relationships between items.

- Works with transaction-style data.
- Produces association rules from discovered item relationships.

Uses

- Market basket analysis
- Shopping recommendations
- Purchase pattern analysis

Applications

- Market basket analysis
  - Shopping recommendations
  - Purchase pattern analysis

Real-Life Example

Real-life example: A supermarket may discover that customers who buy bread often also buy butter.

3.3.2 Basics of Apriori Algorithm

Definition / Explanation

Apriori is a basic algorithm used for association rule mining. It works by finding frequent itemsets in transaction data and then using them to generate association rules. The algorithm uses the idea that if an itemset is not frequent, its larger itemsets containing it will also not be frequent. This helps reduce the number of combinations that need to be examined.

Features

- Finds frequent itemsets.
- Uses support to identify frequent combinations.
- Generates association rules from frequent itemsets.

Uses

- Market basket analysis
- Shopping transactions
- Product relationship analysis

Applications

- Market basket analysis
  - Shopping transactions
  - Product relationship analysis

Real-Life Example

Real-life example: From many shopping bills, Apriori can find frequent combinations such as bread and butter.

3.3.3 Market Basket Analysis

Definition / Explanation

Market Basket Analysis examines customer transaction records to discover products that are frequently purchased together. It commonly uses association rule mining techniques such as Apriori. The discovered relationships can help a business understand purchasing patterns and identify product combinations. For example, a rule may show that customers who purchase bread also frequently purchase butter.

Features

- Analyzes shopping transactions.
- Finds frequently associated products.
- Supports understanding of buying patterns.

Uses

- Retail stores
- Online shopping
- Product recommendation analysis

Applications

- Retail stores
  - Online shopping
  - Product recommendation analysis

Real-Life Example

Real-life example: An online store may find that customers buying a mobile phone often also buy a phone case.

UNIT 4 – Data Warehouse Design

4.1 Star Schema

Definition / Explanation

A Star Schema is a data warehouse design in which a central fact table is directly connected to several dimension tables. The fact table stores measurable business data, while dimension tables provide descriptive information about those facts. The structure looks like a star because the fact table is in the center and dimensions surround it. It is commonly used for simple analytical queries.

Features

- Has a central fact table.
- Connects directly to dimension tables.
- Provides a simple structure for analysis.

Uses

- Sales analysis
- Student performance analysis
- Business reporting

Applications

- Sales analysis
  - Student performance analysis
  - Business reporting

Real-Life Example

Real-life example: A sales warehouse can have a Sales fact table connected to Product, Customer, Date, and Store dimensions.

4.1.1 Snowflake Schema

Definition / Explanation

A Snowflake Schema is a data warehouse design in which dimension tables are further divided into related tables. It is an extension of the star schema. Instead of keeping all descriptive details in one dimension table, related information can be separated into additional tables. The resulting structure is more normalized and appears like a snowflake. It is useful when dimension data has related levels or structures.

Features

- Uses a central fact table.
- Dimensions can be divided into related tables.
- Has a more normalized structure.

Uses

- Student performance data
- Sales analysis
- Hierarchical dimension analysis

Applications

- Student performance data
  - Sales analysis
  - Hierarchical dimension analysis

Real-Life Example

Real-life example: A Product dimension can be separated into Product, Category, and Department tables.

4.1.2 Star and Snowflake Schema

Definition / Explanation

Both Star and Snowflake schemas are used to organize data warehouses for analysis. In a Star Schema, the fact table connects directly to dimension tables. In a Snowflake Schema, dimension tables can be split into additional related tables. Star Schema is simpler to understand, while Snowflake Schema represents more detailed relationships among dimensions. Both can be used for analytical data such as sales or student records.

Features

- Both use fact and dimension concepts.
- Star has simpler direct dimensions.
- Snowflake divides some dimensions into related tables.

Uses

- Sales warehouse
- Student data warehouse
- Reporting systems

Applications

- Sales warehouse
  - Student data warehouse
  - Reporting systems

Real-Life Example

Real-life example: A sales warehouse can use a Star Schema for simple product information or a Snowflake Schema when product categories are stored separately.

4.2 Fact Table and Dimension Table

Definition / Explanation

A fact table stores measurable values related to a business process. Examples of facts include sales amount, quantity, or marks. A dimension table stores descriptive information that gives context to the facts, such as product, customer, date, or student details. In a data warehouse, fact and dimension tables are connected so that users can analyze measurements according to different descriptions.

Features

- Fact table stores measurable values.
- Dimension table stores descriptive details.
- They work together for analysis.

Uses

- Sales facts and product dimensions
- Marks and student dimensions
- Quantity and date dimensions

Applications

- Sales facts and product dimensions
  - Marks and student dimensions
  - Quantity and date dimensions

Real-Life Example

Real-life example: A Sales fact table can store quantity and amount, while a Product dimension stores product name and category.

4.2.2 Relationship Between Fact and Dimension Tables

Definition / Explanation

A fact table is connected to dimension tables through keys. The fact table contains measures and references to dimensions, while dimension tables contain descriptive attributes. This relationship allows users to analyze facts according to different dimensions. For example, sales amount can be analyzed by product, customer, date, or store. The fact table therefore provides measurable events and dimensions provide the context needed to understand them.

Features

- Facts reference dimensions using keys.
- Dimensions describe the facts.
- Together they support multidimensional analysis.

Uses

- Sales by product
- Sales by date
- Marks by student

Applications

- Sales by product
  - Sales by date
  - Marks by student

Real-Life Example

Real-life example: A sales record can reference a Product ID and Date ID so that sales can be analyzed by product and time.

4.3 ETL – Extract, Transform, Load

Definition / Explanation

ETL stands for Extract, Transform, and Load. Extract means collecting data from source systems. Transform means cleaning, converting, organizing, or preparing the data into a suitable form. Load means placing the transformed data into the data warehouse or target system. ETL is important because data may come from different sources and formats. It prepares the data so it can be used for analysis and reporting.

Features

- Extract data from source systems.
- Transform and prepare the data.
- Load prepared data into the warehouse.

Uses

- Sales data integration
- Student data preparation
- Business reporting

Applications

- Sales data integration
  - Student data preparation
  - Business reporting

Real-Life Example

Real-life example: A company extracts sales data from branch systems, cleans and converts it, and loads it into a central warehouse.

4.4 Basic OLAP Operations

Definition / Explanation

OLAP operations are used to analyze data from different views in a data warehouse. The basic operations in the syllabus are Roll-up, Drill-down, Slice, and Dice. Roll-up gives a more summarized view, while Drill-down

moves toward more detailed information. Slice selects a part of the data based on one dimension, and Dice selects a smaller part using multiple dimensions. These operations help users explore analytical data.

Features

- Roll-up summarizes data.
- Drill-down shows more detail.
- Slice and Dice select specific parts of data.

Uses

- Sales analysis
- Time-based reporting
- Product and region analysis

Applications

- Sales analysis
  - Time-based reporting
  - Product and region analysis

Real-Life Example

Real-life example: A manager can roll up daily sales to monthly sales or drill down from yearly sales to a particular day's sales.

UNIT 5 – Tools and Applications

5.1 Introduction to Data Mining Tools

Definition / Explanation

Data mining tools are software applications that help users prepare, analyze, visualize, and mine datasets. The syllabus includes Orange, RapidMiner, and KNIME. These tools can provide visual or workflow-based methods for applying data mining techniques. They can be useful for learning and performing practical tasks such as classification, clustering, visualization, and data workflows.

Features

- Help perform data analysis.
- Support data mining techniques.
- Provide practical workflows or visual interfaces.

Uses

- Classification
- Clustering
- Data visualization and workflows

Applications

- Classification
  - Clustering
  - Data visualization and workflows

Real-Life Example

Real-life example: A student can load a dataset into Orange and create a workflow to perform classification and view the results.

5.1.1 Orange, RapidMiner and KNIME

Definition / Explanation

Orange is an open-source machine learning and data visualization tool and is useful for clustering, classification, and visualization. RapidMiner is a visual drag-and-drop data mining tool that supports machine learning, text mining, and predictive analytics. KNIME Analytics Platform is an open-source platform for data analysis and reporting and is useful for ETL and data workflows. These tools support practical learning and analysis.

Features

- Orange supports visual data mining.
- RapidMiner provides drag-and-drop workflows.
- KNIME supports analysis, reporting, and ETL workflows.

Uses

- Classification
- Clustering
- ETL and data workflows

Applications

- Classification
  - Clustering
  - ETL and data workflows

Real-Life Example

Real-life example: A student can use a visual workflow in one of these tools to load a dataset, apply a technique, and inspect the output.

5.2 Case Studies of Data Mining

5.2.1 Business, Education and Healthcare Case Studies

Definition / Explanation

Data mining case studies show how techniques can be applied to real problems. In business, Market Basket Analysis can be used to study products purchased together. In education, student performance analysis can examine marks and related data. In healthcare, data mining can support disease prediction and patient data analysis. These cases use datasets to discover useful patterns and produce insights for practical decision-making.

Features

- Business – Market Basket Analysis.
- Education – Student performance analysis.
- Healthcare – Disease prediction and patient data analysis.

Uses

- Retail decision-making
- Education analysis
- Healthcare analysis

Applications

- Retail decision-making
  - Education analysis
  - Healthcare analysis

Real-Life Example

Real-life example: A college can analyze student marks to identify performance patterns and areas that may require attention.

5.2.2 Results and Insights from Case Studies

Definition / Explanation

The result of a data mining case study is not only the output of an algorithm but also the useful insight obtained from that output. In a business case, the result may show products often purchased together. In education, it may show patterns in student performance. In healthcare, it may identify patterns related to patient data. The results must be interpreted so that meaningful insights can be understood and used.

Features

- Identify useful patterns from results.
- Interpret the results in the real context.
- Use meaningful insights for decisions.

Uses

- Purchase patterns
- Student performance patterns
- Patient data patterns

Applications

- Purchase patterns
  - Student performance patterns
  - Patient data patterns

Real-Life Example

Real-life example: If analysis shows a strong purchase relationship between two products, a retailer can use that insight when planning product placement.

5.3 Future Scope of Data Mining and Warehousing

Definition / Explanation

Data mining and warehousing have continuing importance because organizations generate large amounts of data and need useful information for decisions. Future applications can involve more advanced analysis, business intelligence, and data-driven systems. Data mining can help discover patterns, while data warehouses can organize information for analysis. As data continues to grow, skills in storing, analyzing, and using data are likely to remain useful in many fields.

Features

- Growing use of data-driven analysis.
- Increasing need for organized analytical data.
- Continued use in business intelligence and decision-making.

Uses

- Business intelligence
- Advanced data analysis
- Data-driven decision-making

Applications

- Business intelligence
  - Advanced data analysis
  - Data-driven decision-making

Real-Life Example

Real-life example: Organizations can combine large historical datasets with mining techniques to understand trends and support future planning.

5.3.1 Emerging Applications

Definition / Explanation

Emerging applications of data mining and warehousing involve using large datasets to understand patterns and support new types of analysis. Possible areas include business intelligence, customer behavior analysis, healthcare data analysis, education analytics, and other data-driven applications. As organizations collect more digital data, methods for storing, organizing, and discovering knowledge from that data become increasingly useful.

Features

- Support new data-driven applications.
- Analyze growing digital datasets.
- Help discover useful patterns and trends.

Uses

- Customer behavior analysis
- Healthcare analytics
- Education analytics

Applications

- Customer behavior analysis
  - Healthcare analytics
  - Education analytics

Real-Life Example

Real-life example: An online service can analyze user activity data to identify changing usage trends.

5.3.2 Role in Decision-Making and Business Intelligence

Definition / Explanation

Data warehousing provides organized data for reporting and analysis, while data mining discovers patterns and useful knowledge from data. Together, they can support business intelligence and decision-making. Organizations can examine historical information, identify trends, and use analytical results to understand business situations. This helps decision-makers use evidence from data instead of depending only on assumptions.

Features

- Provides organized data for analysis.
- Discovers patterns and trends.
- Supports evidence-based decisions.

Uses

- Sales decisions
- Business reporting
- Trend analysis

Applications

- Sales decisions
  - Business reporting
  - Trend analysis

Real-Life Example

Real-life example: A company can analyze historical sales data to understand which products perform better and use the result for planning.

5.3.3 Ethical Considerations

Definition / Explanation

Data mining and warehousing involve collecting, storing, and analyzing data, so ethical considerations are important. Organizations should handle data responsibly and consider privacy, appropriate use, and protection of information. Data should not be used in ways that are unfair or inappropriate. Users and organizations should understand that useful analysis must also respect people and responsible data practices.

Features

- Respect privacy of data.
- Use data for appropriate purposes.
- Protect stored and analyzed information.

Uses

- Customer data handling
- Healthcare data handling
- Student data handling

Applications

- Customer data handling
  - Healthcare data handling
  - Student data handling

Real-Life Example

Real-life example: A healthcare organization should protect patient information when using patient data for analysis.